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PGA-UNet: A Lightweight Prompt-Guided Attention U-Net for Interactive Bone X-Ray Lesion Segmentation

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ABSTRACT Bone X-ray lesion segmentation is a challenging task because lesions are often small, low-contrast, and visually entangled with the surrounding anatomy. In this setting, the main challenge typically lies in localizing the suspicious region rather than refining the lesion's boundary. A practical strategy is to narrow the search space before segmentation itself is performed. We realize this idea through a direct interactive approach: a clinician supplies a coarse bounding box, which the model then uses as a spatial guidance signal.

We propose PGA-UNet, a lightweight prompt-guided CNN that converts the bounding-box prompt into a Gaussian-smoothed plateau heatmap, modulates encoder features according to the prompt through a Prompt Spatial Gate (PSG), and sustains prompt influence in the decoder through a Conditional Attention Decoder (CAD). The goal is not only to use the prompt to indicate where segmentation should occur, but also to keep small-lesion-relevant features active throughout the network.

Attention U-Net serves as the automatic baseline for prompt-free segmentation. At 512×512 resolution under covering prompts, PGA-UNet achieved a Dice score of 0.8788 on BTXRD and 0.8286 on FracAtlas, far exceeding the 0.5160 and 0.3075 obtained by Attention U-Net. Under the same prompt-based setting at 256×256 resolution, PGA-UNet outperformed fine-tuned SAM-Med2D under both covering and off-center prompts, on both datasets, with both models trained on the same 50/50 mixture of covering and off-center prompts. The largest margin appeared on the 50 smallest lesions in BTXRD and FracAtlas, where SAM-Med2D's Dice score dropped to 0.1903 while PGA-UNet reached 0.7650; center-based localization (CBL) likewise favored PGA-UNet (0.9261 versus 0.4489), indicating substantially better lesion localization.

Component-contribution analysis shows that PSG and CAD are most effective when combined, and that the benefit of the Gaussian prompt over a binary box only becomes apparent within the full architecture. With 2.95 million parameters, PGA-UNet reduces parameter count, FLOPs, and GPU inference latency by approximately $92\times$, $11.9\times$, and $6.0\times$, respectively, relative to SAM-Med2D in the tested 256×256 setting. These results suggest that, for very small lesions, a prompt is most effective when treated as a dense spatial prior that remains active from early encoding through final decoding.

INDEX TERMS bone X-ray, interactive segmentation, medical image segmentation, prompt-guided learning, Attention U-Net

I. INTRODUCTION

BONE X-ray imaging is widely used in screening, diagnostic support, and follow-up because of its speed, low cost, and broad accessibility. Lesion segmentation in radiographs, however, remains difficult when the lesion is very small, low-contrast, or partially obscured by surrounding anatomy. In these cases, one of the main challenges lies in localization: before any boundary can be refined, the model must first identify the suspicious region within the entire image.

This is why small-lesion segmentation is rarely just a boundary-delineation problem. When a lesion occupies only a tiny fraction of the image, searching the entire radiograph is both inefficient and error-prone. A natural response is to narrow the search space first, then segment within the reduced region. Fully automatic methods attempt to learn this process internally. Although U-Net and Attention U-Net [3], [4] remain representative models of this approach, they still require the network to localize and segment the lesion entirely on its own, without any external assistance.

An alternative is to draw region-of-interest guidance from an external source. In clinical practice, a clinician can easily draw a coarse bounding box around a suspicious area, something a fully automatic model can rarely detect reliably for very small lesions. In this sense, a bounding-box prompt is more than an interactive convenience; it is a practical means of injecting spatial knowledge into the pipeline before detailed segmentation begins. Prompt-based interactive segmentation methods have grown out of this idea, from clicks and scribbles [5], [6] to recent promptable foundation models such as SAM and SAM-Med2D [1], [2].

For the present task, however, one issue remains unresolved. Once a prompt has identified the region of interest, should it serve merely as a positional constraint for mask generation, or should it continue to shape feature extraction so that faint lesion signals survive downsampling? This distinction matters most for very small lesions, where a small spatial deviation can substantially affect the segmentation outcome.

We address this question with PGA-UNet, a lightweight prompt-guided CNN in which the box prompt is treated as a structured spatial prior rather than an auxiliary input channel. The prompt is encoded as a plateau-style Gaussian heatmap, which is then modulated into encoder features through a Prompt Spatial Gate (PSG) and reused in the decoder through a Conditional Attention Decoder (CAD). This design is intended to keep prompt-guided lesion features active throughout the network, rather than merely confining segmentation to a predefined box region.

This perspective also shapes how the experiments are interpreted. Because Attention U-Net receives no spatial guidance from a clinician, its role is only to show how difficult the problem becomes when localization must be fully automatic. SAM-Med2D, in contrast, is chosen as the direct comparison because it receives the same bounding-box prompt input. The central question here is therefore not whether prompts are useful in general, but whether a lightweight architecture can exploit a coarse clinician prompt more effectively by preserving weak lesion signals, rather than treating the prompt merely as a spatial constraint.

The main contributions of this article are as follows.

- We propose PGA-UNet, a lightweight prompt-guided architecture for interactive bone X-ray lesion segmentation.
- We design an effective prompt-conditioning mechanism that combines a plateau-style Gaussian prompt, an encoder-side Prompt Spatial Gate (PSG), and a decoder-side Conditional Attention Decoder (CAD).
- We formulate an offline training and online interactive inference framework, with explicit mathematical definitions for prompt encoding, encoder gating, and decoder conditioning.
- We evaluate the method on the BTXRD and FracAtlas datasets under covering and off-center prompt conditions, comparing against Attention U-Net and SAM-Med2D using Dice, a center-based localization score (CBL), and HD95, together with Monte Carlo cross-validation and computational efficiency measurements.
- We report component-contribution analysis, efficiency evaluation, and small-lesion analysis that clarify where the proposed prompt-conditioning design provides the greatest benefit.

II. RELATED WORK

Small-lesion segmentation in medical imaging has long been constrained by a simple practical fact: before a model can segment a tiny target well, it must first narrow down where to look. In the fully automatic route, this search-space tightening is learned implicitly inside the network. U-Net established the canonical multi-scale encoder-decoder template for biomedical segmentation [3], and Attention U-Net later introduced attention gates on skip connections so that decoder-side fusion could suppress irrelevant spatial responses more selectively [4]. These models improved automatic segmentation quality, but they still assume that the network itself must localize and segment the lesion from the raw image alone.

Interactive medical segmentation developed in parallel from the observation that partial user guidance can simplify the problem substantially [5], [6]. Early methods used clicks, scribbles, and geodesic cues; later work incorporated boxes as practical coarse prompts. Conceptually, these methods move part of the localization burden from the model to the user. For tiny radiographic lesions, that shift is especially meaningful: a clinician may be able to indicate a suspicious region coarsely even when a fully automatic system would struggle to find it reliably in the whole image.

The next major step was promptable foundation models such as SAM, SAM-Med2D, and MedSAM, which generalized prompt-driven segmentation across many medical domains [1], [2], [9]. These models are flexible, but their prompt usage is architecturally different from lightweight task-specific CNNs, and their computational cost is much higher. More importantly for the present task, they still leave open a design question that becomes acute for very small radiographic lesions: once a prompt has already localized the approximate region, should the prompt merely indicate where the mask should be generated, or should it continue to shape feature extraction and decoding so that weak lesion evidence is not lost?

The present work is motivated by that gap. In grayscale bone radiographs, the problem is not only to know *where* the lesion roughly is, but also to preserve weak lesion evidence while suppressing surrounding clutter after the search space has already been tightened. A lightweight prompt-guided CNN may therefore benefit from making prompt information active from early encoding to late decoding instead of using the prompt only as an initial cue or positional restriction.

TABLE 1. Representative prior work and the remaining gap addressed in this article.

Work	Principle	Method	Performance Evidence	Pros	Cons
U-Net [3]	Fully convolutional encoder-decoder	Multi-scale skip-connected segmentation network	Broad biomedical benchmarks; Dice-based overlap reporting	Simple, efficient, strong canonical baseline	No prompt guidance; all localization must be automatic
Attention U-Net [4]	Skip-feature filtering by attention gates	U-Net with gated skip connections	Medical image segmentation tasks with overlap metrics such as Dice and IoU	Stronger automatic baseline than plain U-Net	Attention is still learned without explicit prompt conditioning
DeepIGeoS and related interactive methods [6]	User guidance reduces search space	Geodesic or interaction-aware segmentation	Interactive medical segmentation under user input	Better reflects clinical interaction	Interaction mechanism can be task-specific or expensive to maintain
SAM-Med2D / MedSAM [2], [9]	Promptable foundation segmentation	Large pretrained model with prompt-driven mask decoding	Multi-dataset medical evaluation with prompt-based metrics	Flexible and broadly applicable	Heavy; prompt influence is not designed as a lightweight end-to-end radiograph-specific prior
This work	Prompt as a dense spatial prior throughout the network	Gaussian plateau prompt + PSG + CAD in a lightweight CNN	BTXRD and FracAtlas; Dice, IoU, HD95, CBL, efficiency, small-lesion analysis	Lightweight, prompt-aware from encoder to decoder, robust to prompt perturbation	Still requires supervised masks and fixed-resolution preprocessing

III. METHOD

A. OFFLINE AND ONLINE FRAMEWORK

The proposed system has two stages. In the offline stage, the model is trained on radiographs paired with lesion polygons. Each polygon is converted into a training prompt by generating a simulated bounding box and then transforming that box into a Gaussian-smoothed plateau heatmap. The network learns to map the image-prompt pair to the binary lesion mask. In the online stage, a user provides a coarse bounding box around a suspected lesion on a new radiograph. The same prompt-construction procedure is applied, and the trained model predicts the final lesion mask.

The scientific contribution of this framework is that the prompt is not treated as an optional side input used only at the first layer. Instead, the prompt is encoded as a dense spatial prior and made active at both encoder and decoder stages. The practical consequence is improved robustness when the prompt is coarse, mildly off-center, or applied to a very small lesion.

B. PROBLEM SETTING

Given a grayscale radiograph $\mathbf{I} \in \mathbb{R}^{H\times W}$ and a user-provided bounding box $\mathbf{B}$, the goal is to predict a binary lesion mask $\hat{\mathbf{M}}$. Let $\mathbf{H}(\mathbf{B})$ be the prompt heatmap derived from the box. PGA-UNet predicts

$$\hat{\mathbf{M}} = \mathbf{1}\left[f_{\mathrm{PGA}}\left(\mathbf{I}, \mathbf{H}(\mathbf{B}); \theta\right) \geq 0.5\right]. \tag{1}$$

C. PROMPT REPRESENTATION

The input prompt is not encoded as a hard binary rectangle. Instead, the box interior is rasterized into a plateau map at the original image resolution and its boundaries are softened with a fixed 31×31 Gaussian kernel, before the heatmap is resized and padded down to the target square resolution together with the image. The kernel size is not rescaled with the target resolution (256×256 or 512×512): applying it once at the original resolution, prior to downsampling, keeps the effective blur consistent relative to whichever square frame the network ultimately sees. The plateau covers exactly the (possibly expanded) box with no additional minimum margin; coverage of the region of interest is guaranteed by the box expansion described below, not by a separate margin step.

During evaluation, prompts are tested in two settings. A covering prompt expands the lesion bounding box by 15%–45% of lesion width and height. An off-center prompt first applies the same expansion and then shifts the box by up to 30% of lesion width and height while still covering the lesion. A single model is trained per dataset and resolution, using a random 50/50 mixture of the covering and off-center prompt conditions at training time; the two conditions are then evaluated separately at test time, unless otherwise stated in the ablation study.

Let (x_1, y_1, x_2, y_2) denote the tight lesion box and let $w = x_2 - x_1$ and $h = y_2 - y_1$. The simulated covering box expands each side by a random fraction in $[0.15, 0.45]$ of w or h during training and by a fixed fraction of 0.30 during evaluation. The off-center box applies the same expansion and then adds a translation bounded by $0.30w$ horizontally and $0.30h$ vertically while preserving lesion coverage. This simulation is not a substitute for radiologist-drawn prompts, but it provides a controlled way to test sensitivity to plausible box variation.

Algorithmically, prompt construction follows three steps: 1) derive a lesion box from the polygon annotation or user input and expand it as described above, 2) rasterize the expanded box into a plateau mask at the original image resolution, and 3) smooth the plateau boundary with the fixed Gaussian kernel before resizing to the target resolution. This sequence avoids the hard-edge dependence of a binary box while preserving a clear spatial prior.

D. PROMPT SPATIAL GATE

Let $\mathbf{x}^l$ denote the encoder feature map at level l. PSG modulates it with a prompt-derived attention map $\mathbf{A}^l_{\mathrm{PSG}}$:

$$\tilde{\mathbf{x}}^l = \mathbf{x}^l \odot \left(1 + \alpha^l_{\mathrm{PSG}} \mathbf{A}^l_{\mathrm{PSG}}\right), \tag{2}$$

where $\odot$ is element-wise multiplication and α^l_{PSG} is learnable. In the implementation, α^l_{PSG} is initialized to 0.1 and clamped to $[0, 1]$. The residual form keeps the original feature stream available while emphasizing prompt-relevant regions.

E. CONDITIONAL ATTENTION DECODER

CAD extends decoder attention by conditioning the skip gate on prompt-encoded features. Let $\mathbf{g}^l$ be the decoder gating feature and $\mathbf{p}^l_{\mathrm{enc}}$ the prompt encoding at the same scale. The conditioned gate is

$$\mathbf{g}'^{\,l} = \mathbf{g}^l + c^l \alpha^l_{\mathrm{CAD}} w_l \mathbf{p}^l_{\mathrm{enc}}, \tag{3}$$

where c^l is a prompt confidence scalar, α^l_{CAD} is learnable, and w_l is a fixed depth coefficient. This design helps the decoder remain aligned with the prompt even when the box is not perfectly centered.

The prompt encoder in CAD uses two 3×3 convolutions with instance normalization and ReLU. The prompt-confidence scalar c^l is produced by global average pooling followed by a 1×1 convolution and a sigmoid nonlinearity. The decoder depth coefficients are fixed to $\{1.0, 0.7, 0.4, 0.2\}$

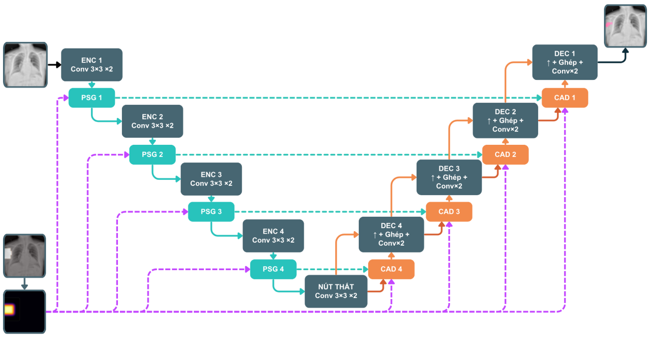

FIGURE 1. Overview of PGA-UNet. Prompt information derived from the bounding box is injected into encoder features through PSG and reused in decoder skip attention through CAD.

from deep to shallow layers. With feature scale 4, the channel widths are $\{16, 32, 64, 128, 256\}$ from the shallowest encoder stage to the bottleneck. The CAD mixing coefficient is parameterized as a sigmoid-transformed scalar and initialized near 0.3.

Taken together, the method can be summarized as follows: 1) preprocess the image and prompt into a common square resolution, 2) encode prompt-relevant spatial priors with PSG in the encoder, 3) propagate prompt-conditioned context through CAD in the decoder, and 4) predict the final binary lesion mask. The scientific meaning of the contribution is that prompt information is both spatially softened and repeatedly reused, rather than being injected once and left to dissipate.

IV. EXPERIMENTAL SETUP

A. DATASETS

We evaluate on two public bone radiograph datasets. BTXRD is a primary bone tumor X-ray dataset with polygon annotations for classification, localization, and segmentation [7]. Our BTXRD split contains 1,493/187/187 images with 1,848/238/232 lesion polygons for training, validation, and testing, respectively. FracAtlas is a fracture radiograph dataset with classification, localization, and segmentation labels [8]. After data cleaning and polygon normalization, our FracAtlas split contains 573/72/72 images with 730/95/92 lesion polygons for training, validation, and testing, respectively.

Splits are performed at the image level, and all polygons from the same image are kept in the same split. The public metadata were insufficient to verify strict patient-level separation. Each polygon is treated as one promptable lesion sample during training. For image-level evaluation, if an image contains multiple lesions, the model is run once per lesion box and the probability maps are merged by pixelwise maximum before thresholding at 0.5. Ground-truth masks are merged by union in the same image.

Inputs are resized isotropically and padded to square resolutions of 256×256 or 512×512 to preserve aspect ratio. Images, masks, and prompt maps undergo the same geometric preprocessing.

B. TRAINING DETAILS

All CNN models were implemented in PyTorch and trained on an NVIDIA Tesla T4 GPU with 16 GB memory. PGA-UNet and Attention U-Net were trained separately on each dataset and resolution. In other words, the article studies one shared architecture family instantiated as separate models with separate parameter sets for BTXRD and FracAtlas, rather than a single jointly trained cross-dataset model. The optimizer was AdamW with learning rate 10^{-4} and weight decay 10^{-4}. The loss was the sum of binary cross-entropy and Dice loss. Batch size was 4, the maximum training budget was 100 epochs, and early stopping with patience 15 was used. The primary model-selection criterion was validation Dice under the covering-prompt condition.

Training-time augmentation included horizontal flipping and random rotation within $\pm 15^{\circ}$. The implementation also applied prompt dropout or prompt noise to a subset of training iterations to reduce over-reliance on an ideal prompt. PGA-UNet is trained on a random 50/50 mixture of the covering-prompt (zoom-out) and off-center (shift) conditions, so that both conditions are represented during training rather than only at evaluation time. The current article should therefore be read as a first controlled study under simulated box prompts, not yet as an evaluation under fully realistic radiologist-drawn boxes. For the fine-tuned SAM-Med2D comparison, the pretrained image encoder was frozen except for its lightweight Adapter layers, while the prompt encoder and mask decoder were fully updated on each dataset under the same split protocol. Rather than the original authors' training-time box perturbation, a tight box with a few pixels of random jitter combined with iterative point-based refinement, SAM-Med2D was finetuned box-only, with each training box independently drawn on a per-sample 50/50 basis from the same covering-prompt protocol used for PGA-UNet: a random expansion of 15%–45% per side, or the same expansion followed by an off-center shift of up to 30%. PGA-UNet and SAM-Med2D are therefore trained on the same 50/50 mixture of covering and off-center conditions, so neither model is put at a training-distribution disadvantage when evaluated under off-center prompts at test time, and the comparison isolates architectural differences rather than an unrelated train-test distribution shift. Validation during SAM-Med2D finetuning and the final test split used the same covering-prompt protocol as PGA-UNet, with batch size 4, a maximum of 100 epochs, early stopping with patience 30, and learning rate 10^{-5}. Additional stability experiments were performed using Monte Carlo cross-validation, that is, repeated random image-level splits rather than strict non-overlapping k-fold partitioning.

Two evaluation details are important for interpreting the results. First, prompt generation itself is resolution-agnostic: the plateau heatmap is built and smoothed with a fixed 31×31 Gaussian kernel at the original image resolution, and only the resulting heatmap is resized and padded down to 256×256

or 512×512 together with the image, so the same absolute blur is compressed differently by the two target resolutions rather than being parameterized separately for each. Second, the Monte Carlo cross-validation experiments are auxiliary stability evidence rather than replacements for the fixed train/validation/test split used in the main tables.

C. BASELINES AND METRICS

PGA-UNet is compared with Attention U-Net [4] as the main automatic baseline and with SAM-Med2D [2] as the main prompt-based baseline. Attention U-Net is trained without prompts and is used to characterize how difficult the task is when localization must be solved automatically. SAM-Med2D is evaluated at 256×256 in fine-tuned form with the same dataset splits and the same box prompts as PGA-UNet, making it the primary prompt-matched comparison.

To further isolate whether PGA-UNet's advantage comes from its architecture or simply from having access to the box prompt at all, we add two prompt-matched conventional baselines built on the same Attention U-Net backbone, given the identical box prompt PGA-UNet receives but without PGA-UNet's Gaussian-prior heatmap, Prompt Spatial Gate, or Conditional Attention Decoder. Attention U-Net with a concatenated prompt channel appends the prompt heatmap as a second input channel and predicts on the full image through a plain skip-connection decoder. Attention U-Net on prompt crops instead restricts the network's field of view: the input image is cropped to the prompt box before resizing to the model's input resolution, the plain Attention U-Net predicts a mask on that crop alone, and the prediction is pasted back into the full frame for evaluation. Both baselines are trained under the same covering and off-center box protocol as PGA-UNet, so any remaining gap reflects the architecture rather than an unequal prompt.

We report Dice, IoU, precision, recall, HD95, and center-based localization (CBL). For a predicted mask centroid (x_p, y_p), a ground-truth centroid (x_g, y_g), and the diagonal length D_{GT} of the ground-truth bounding box, CBL is defined as

$$\mathrm{CBL} = \max\left(0, 1 - \frac{\sqrt{(x_p - x_g)^2 + (y_p - y_g)^2}}{D_{\mathrm{GT}}}\right). \quad (4)$$

HD95 is computed from bidirectional boundary distances. If both masks are empty, HD95 is set to 0. If exactly one mask is empty, HD95 is set to the image side length (256 or 512 pixels). No samples are discarded.

D. SCOPE OF ANALYSIS

This article focuses on prompt-guided segmentation itself and excludes the auxiliary screening pipeline explored separately in the thesis manuscript. To keep the article compact, we retain only the subgroup analyses that most directly support the main claim: the role of prompt-guided localization, robustness under imperfect prompts, and effectiveness on very small lesions.

TABLE 2. PGA-UNet Dice under covering and off-center prompts at two resolutions.

Dataset	Resolution	Covering	Off-center
BTXRD	256×256	0.8449	0.8111
BTXRD	512×512	**0.8788**	**0.8496**
FracAtlas	256×256	0.7974	0.7619
FracAtlas	512×512	**0.8286**	**0.7925**

V. RESULTS

A. MAIN COMPARISON ACROSS DATASETS AND RESOLUTIONS

This first group of experiments addresses the most basic question: does PGA-UNet remain effective across both datasets, across both tested input resolutions, and across both prompt scenarios? These experiments primarily validate the main system-level contribution, namely that prompt-conditioned processing through both encoder and decoder remains useful after the search space has already been tightened by a coarse box prompt. PGA-UNet performs well on both BTXRD and FracAtlas under the intended workflow with coarse box guidance. Across the main experiments, the model remains competitive at both 256×256 and 512×512 and under both covering and off-center prompt settings. The consistent trend across datasets, resolutions, and prompt scenarios is that prompt information continues to help even though BTXRD and FracAtlas differ substantially in lesion appearance and are trained as separate model instances rather than through one joint cross-dataset model.

B. EFFECT OF INPUT RESOLUTION

This experiment isolates the contribution of image resolution within the same architecture and prompt protocol. It supports the claim that the proposed method is especially useful when small-lesion evidence would otherwise be weakened after the search space has been reduced but the lesion signal itself is still fragile. PGA-UNet was trained independently at 256×256 and 512×512 on each dataset. The higher resolution consistently improved Dice under both prompt conditions, and the gain was slightly larger on BTXRD than on FracAtlas. This difference is consistent with the datasets themselves: BTXRD contains many small or irregular lesions whose local evidence degrades noticeably after aggressive downsampling, whereas FracAtlas more often contains elongated but visually simpler fracture structures.

The resolution comparison should not be overinterpreted as a pure image-size effect because the prompt heatmap is built once at the original image resolution with a fixed Gaussian kernel and only then downsampled together with the image, so the same absolute blur is compressed more aggressively into the 256×256 frame than into the 512×512 one. Even with that caveat, the main trend remains useful: prompt-guided segmentation of small radiographic lesions

TABLE 3. Monte Carlo cross-validation for PGA-UNet at 512 $\times$ 512 (4 repeated random splits, mean $\pm$ standard deviation).

Dataset	Prompt	Dice	CBL	HD95
BTXRD	Covering	0.8715 ± 0.0083	0.9600 ± 0.0028	8.09 ± 0.67
	Off-center	0.8357 ± 0.0120	0.9390 ± 0.0062	10.45 ± 0.65
FracAtlas	Covering	0.8281 ± 0.0052	0.9597 ± 0.0039	6.93 ± 0.08
	Off-center	0.7862 ± 0.0114	0.9297 ± 0.0054	8.75 ± 0.48

benefits from preserving more spatial detail, especially when the lesion occupies only a tiny fraction of the image.

C. MONTE CARLO CROSS-VALIDATION

This experiment examines whether the main findings remain stable when the image-level split is redrawn repeatedly under limited data. It supports the robustness claim at the dataset-partition level rather than at the prompt-perturbation level. Supplementary experiments therefore used Monte Carlo cross-validation, that is, four repeated random image-level splits of the 512 $\times$ 512 PGA-UNet configuration, evaluated separately under covering and off-center prompts. These results are not intended to replace the fixed-split main tables, but they provide additional evidence that the prompt-guided gains are not tied to a single favorable split.

The standard deviation across the four repeated splits stays under 0.012 Dice in every row, and the covering-versus-off-center ordering established in the main split (Table 2) is preserved in every individual fold. This indicates that the resolution and prompt-condition trends reported above are not an artifact of a single favorable train/validation/test partition.

D. COMPARISON WITH THE AUTOMATIC BASELINE

Attention U-Net receives no clinician-provided box, so this comparison is not a like-for-like test of two prompt-guided methods. Its purpose is instead to measure how difficult this segmentation task is, over the full test set, when the network must localize the lesion entirely on its own, which is exactly the step that a coarse clinician prompt is meant to remove. Table 4 compares PGA-UNet with Attention U-Net at 512 $\times$ 512 under this asymmetric information setting. On both BTXRD and FracAtlas, Attention U-Net's low Dice together with its very high HD95 shows that localization, not boundary refinement, is the main bottleneck without a prompt; its CBL is also low (0.4658–0.5995), confirming that the predicted region is often not even centered on the lesion. The resulting gap to PGA-UNet is therefore evidence that clinician-provided spatial guidance is essential for these very small lesions, not a claim that the PGA-UNet architecture is inherently superior to Attention U-Net.

E. ATTENTION U-NET TOP- AND BOTTOM-DICE SUBSETS

This subgroup analysis is an extension of the automatic-baseline comparison. After comparing PGA-UNet with Attention U-Net at the dataset level, we ask a more specific question: how does PGA-UNet behave on images where the automatic baseline is relatively strong versus relatively weak? The purpose is to clarify the practical meaning of the Attention U-Net comparison rather than to define objective data difficulty. We therefore compare Attention U-Net top-Dice and bottom-Dice image groups on both BTXRD and FracAtlas. Specifically, for each dataset we select the 50 test images with the highest Attention U-Net Dice scores and the 50 test images with the lowest Attention U-Net Dice scores, then re-evaluate PGA-UNet on the same two groups. This analysis does not define intrinsic data difficulty; it only reports performance on subsets induced by the Attention U-Net ranking.

On BTXRD, PGA-UNet improves over Attention U-Net even in the top-Dice group, raising Dice from 0.8629 to 0.9036. The contrast becomes much larger in the bottom-Dice group, where Attention U-Net collapses to 0.0335 Dice while PGA-UNet still reaches 0.8521. FracAtlas shows the same

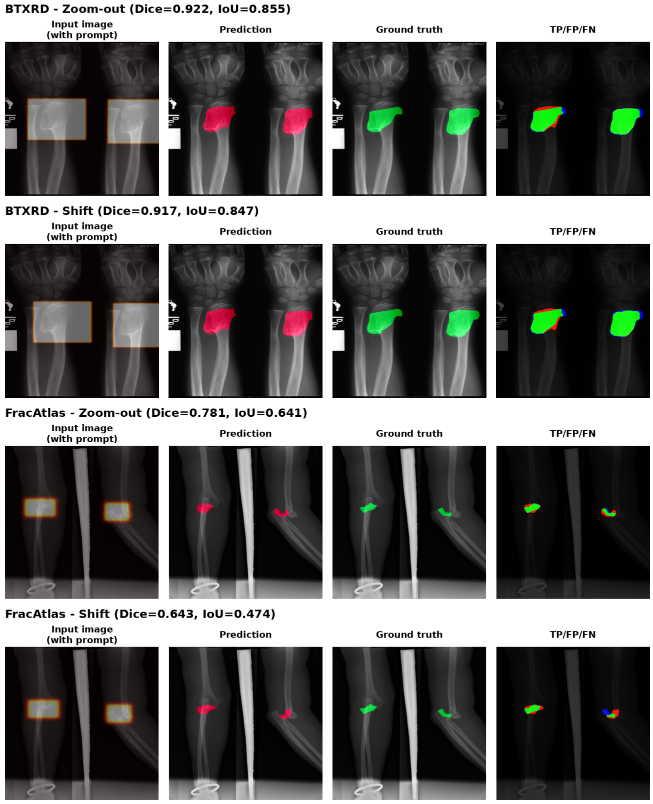

FIGURE 2. Qualitative behavior under prompt perturbation. PGA-UNet remains more stable when the prompt is shifted away from the ideal covering box, although thin fracture patterns are still harder than larger compact lesions.

TABLE 4. Comparison with the automatic baseline at 512 $\times$ 512 under covering prompts.

Dataset	Model	Dice	CBL	HD95
BTXRD	Attention U-Net	0.5160	0.5995	129.67
	PGA-UNet	**0.8788**	**0.9619**	**9.08**
FracAtlas	Attention U-Net	0.3075	0.4658	110.83
	PGA-UNet	**0.8286**	**0.9615**	**6.21**

TABLE 5. **Top-Dice and bottom-Dice comparison under covering prompts at 512 × 512 (50 images per group).**

Dataset	Group	Model	Dice	CBL	HD95
BTXRD	Top-Dice	Attention U-Net	0.8629	0.9436	39.32
		PGA-UNet	**0.9036**	**0.9709**	**9.23**
	Bottom-Dice	Attention U-Net	0.0335	0.0928	270.72
		PGA-UNet	**0.8521**	**0.9459**	**5.19**
FracAtlas	Top-Dice	Attention U-Net	0.4428	0.6277	82.08
		PGA-UNet	**0.8318**	**0.9666**	**6.37**
	Bottom-Dice	Attention U-Net	0.1529	0.3345	135.76
		PGA-UNet	**0.8151**	**0.9582**	**6.25**

qualitative pattern with a different baseline severity: even the Attention U-Net top-Dice group only reaches 0.4428, well below its BTXRD counterpart, which indicates that the automatic baseline struggles more broadly on FracAtlas; PGA-UNet nonetheless reaches 0.8318 on that same group and remains close to that level, at 0.8151, on the bottom-Dice group. Because the four groups are defined directly from Attention U-Net scores, these values should not be interpreted as global averages; instead, they highlight how PGA-UNet behaves on cases where the automatic baseline is relatively strong or weak.

The qualitative examples help clarify this behavior. In the Attention U-Net top-Dice groups, the automatic baseline can often place the predicted mask in roughly the correct anatomical region, but it may still produce spurious responses far from the lesion. Even when the lesion itself is covered reasonably well, those distant false positives reduce overlap-based metrics such as Dice and IoU, which is consistent with the still-elevated HD95 values (39.32 on BTXRD, 82.08 on FracAtlas) despite a moderate-to-low Dice. PGA-UNet is less prone to this pattern because the bounding box prompt constrains the model to a more relevant spatial region. In the bottom-Dice groups, the failure is more severe: the automatic baseline may localize the wrong region or miss the lesion almost entirely. A likely reason is that the attention gate itself has no guarantee of learning a clinically correct gating policy: it filters whatever features it has learned to emphasize, whether those features correspond to lesions or to spurious high-energy structures. When the learned response is dominated by background clutter or artifact-like signals with strength comparable to, or larger than, the true lesion evidence, the gate can no longer rescue localization. In practice, it may suppress only mild background activity while remaining ineffective against stronger false responses. In contrast, PGA-UNet receives a coarse spatial prior and can therefore identify the lesion region more reliably before refining the mask.

F. MATCHED COMPARISON WITH SAM-MED2D

This comparison tests the architectural contribution of the proposed prompt handling against a prompt-based foundation baseline under matched input conditions. Both models receive the same prompt type and the same resolution, so the difference is more directly tied to how prompt information is represented and propagated after localization has already been externally provided. Table 6 reports Dice at 256×256 under both covering and off-center prompts. PGA-UNet achieved higher Dice than fine-tuned SAM-Med2D under both prompt conditions on both datasets. The small-lesion analysis presented next should be read as an extension of this SAM-Med2D comparison to the setting where prompt usage is expected to matter most.

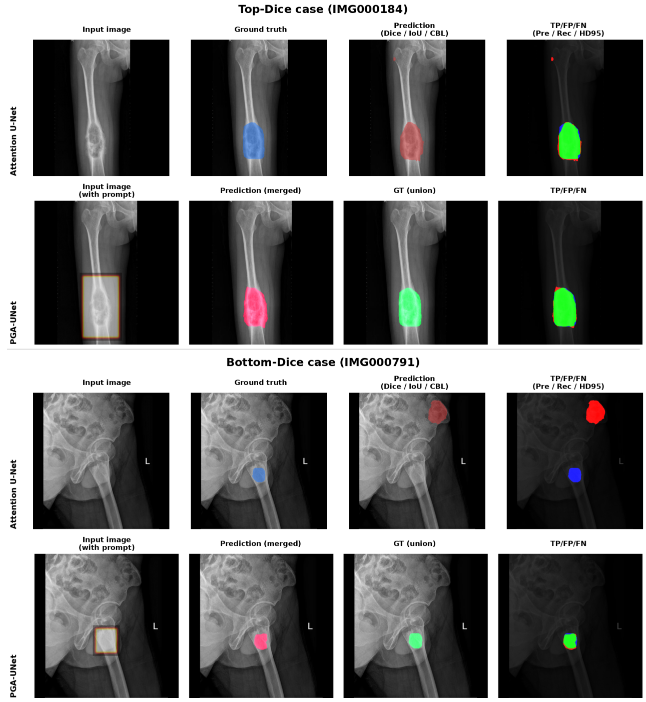

FIGURE 3. **Qualitative comparison between Attention U-Net and PGA-UNet on BTXRD for one case from the Attention U-Net top-Dice group (IMG000184) and one case from the Attention U-Net bottom-Dice group (IMG000791). Each row shows four panels; for PGA-UNet, the input-image and box-prompt panels are merged into a single "Input image (with prompt)" panel.**

TABLE 6. **Dice comparison with fine-tuned SAM-Med2D at 256 × 256.**

Dataset	Prompt	PGA-UNet	SAM-Med2D-FT
BTXRD	Covering	**0.8449**	0.7624
	Off-center	**0.8111**	0.7542
FracAtlas	Covering	**0.7974**	0.7439
	Off-center	**0.7619**	0.7275

Under covering prompts, PGA-UNet outperforms fine-tuned SAM-Med2D by 0.0825 Dice on BTXRD and 0.0535 Dice on FracAtlas at the same 256×256 resolution; under off-center prompts the margin is 0.0569 and 0.0344 Dice, respectively. PGA-UNet therefore remains ahead in absolute Dice under both prompt conditions on both datasets, indicating that once the approximate lesion region is already known, its way of using prompt information remains more effective in this setting.

The relative degradation from covering to off-center

TABLE 7. Dice, CBL, and HD95 on the 50 smallest lesions per dataset. IoU is omitted here because it is a noisy, boundary-sensitive statistic at this lesion size; Dice, CBL, and HD95 give a more reliable picture of whether the model both finds and covers the correct region.

Dataset	Model	Dice	CBL	HD95
BTXRD	SAM-Med2D-FT 256 × 256	0.1903	0.4489	240.72
	PGA-UNet 256 × 256	0.7650	0.9261	3.11
	PGA-UNet 512 × 512	**0.8483**	**0.9451**	**2.32**
FracAtlas	SAM-Med2D-FT 256 × 256	0.3736	0.7116	105.19
	PGA-UNet 256 × 256	0.7973	0.9569	5.62
	PGA-UNet 512 × 512	**0.8348**	**0.9609**	**4.83**

prompts is worth reading carefully rather than as a simple robustness ranking. PGA-UNet's own drop is 0.0338 Dice on BTXRD and 0.0355 Dice on FracAtlas (Table 2), while fine-tuned SAM-Med2D's drop is smaller, 0.0082 and 0.0164 Dice, respectively. As described in Section IV-B, both models are trained on the same 50/50 mixture of covering and off-center prompts, so this difference is not explained by one model seeing a prompt condition the other did not; it instead reflects how each architecture handles an off-center prompt at test time. PGA-UNet nonetheless still achieves higher absolute Dice than fine-tuned SAM-Med2D under off-center prompts on both datasets, consistent with the view that PGA-UNet's off-center behavior comes from how PSG and CAD propagate and reuse prompt information architecturally.

G. PERFORMANCE ON SMALL LESIONS

This experiment is an extension of the SAM-Med2D comparison to the setting in which the contribution of prompt-aware feature preservation should be most visible, namely very small lesions. To keep subgroup analysis objective, we retain only the small-lesion subset because it is defined directly by lesion size rather than by a model-dependent ranking rule. For each dataset, the subset is formed by selecting the 50 test images with the smallest total lesion area, evaluated under covering prompts. This subset is then used for two related comparisons: PGA-UNet versus fine-tuned SAM-Med2D at the matched resolution of 256 × 256, and PGA-UNet$_{256}$ versus PGA-UNet$_{512}$ to examine the effect of resolution. All metrics are computed at a shared 512 × 512 reference frame, with the 256 × 256 predictions of SAM-Med2D and PGA-UNet upsampled before scoring, so that the three models are compared on identical ground-truth masks.

On the BTXRD small-lesion subset, PGA-UNet at 256 × 256 improves Dice by 0.5747 over fine-tuned SAM-Med2D at the same resolution, and increasing the PGA-UNet input size to 512 × 512 yields a further gain of 0.0833 Dice. The HD95 gap is even more striking: fine-tuned SAM-Med2D averages 240.72 pixels, close to the maximum possible penalty on a 512 × 512 frame, indicating that it frequently misses the small lesion entirely rather than producing a merely imprecise boundary. CBL confirms this from an independent angle: fine-tuned SAM-Med2D's predicted centroid drifts far from the true lesion centroid (CBL of 0.4489 on BTXRD and 0.7116 on FracAtlas), whereas PGA-UNet keeps CBL between 0.9261 and 0.9609 across both resolutions and datasets, meaning the predicted region stays centered on the lesion even when the lesion is only a few pixels wide. On the FracAtlas small-lesion subset the same pattern holds at a smaller magnitude: PGA-UNet at 256×256 improves Dice by 0.4237 over fine-tuned SAM-Med2D, with a further 0.0375 Dice gain at 512 × 512. These results are consistent with the view that small lesions benefit from both explicit prompt conditioning and higher spatial resolution.

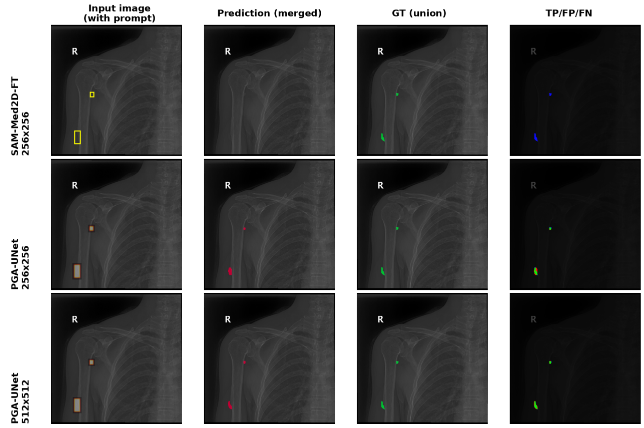

FIGURE 4. Qualitative comparison on small lesions in BTXRD (IMG000868, two small polygons). The examples contrast fine-tuned SAM-Med2D at 256 × 256, PGA-UNet at 256 × 256, and PGA-UNet at 512 × 512; the input-image and box-prompt panels are merged into a single "Input image (with prompt)" panel. SAM-Med2D produces no visible prediction from the same prompt (Dice 0.000), while both PGA-UNet variants recover both lesions.

One plausible explanation is that small lesions already provide limited visual evidence, and resizing a large radiograph to 256 × 256 weakens that evidence further. Under this condition, how prompt information is used becomes especially important. In SAM-Med2D, the box prompt primarily acts as a coarse positional cue for mask generation. When the lesion occupies very few pixels after resizing, this cue alone may be insufficient to recover a detailed mask. By contrast, PGA-UNet uses the bounding box as a dense spatial prior that amplifies the lesion region and propagates prompt information through both the encoder and decoder. This design appears better suited to preserving weak local evidence for small lesions. The stronger result of PGA-UNet at 512×512 is also consistent with this interpretation, since higher resolution preserves more lesion detail before prompt-guided decoding.

H. EFFICIENCY ANALYSIS

This experiment evaluates whether the proposed gains are achieved with a practical computational profile. It supports the claim that the method is suitable for repeated low-latency interactive use rather than only for offline batch inference. Table 8 reports the efficiency comparison, measured on the same NVIDIA Tesla T4 GPU used for training (mean latency over repeated forward passes, batch size 1). PGA-UNet contains only 2.95M parameters. In the tested 256 × 256 setting, this corresponds to an approximately 92-fold reduction in

TABLE 8. Efficiency comparison in the tested environment.

Model	Params	GFLOPs	Size	GPU ms	CPU ms
PGA-UNet 256	2.95M	7.74	11.35 MB	8.17	109.17
PGA-UNet 512	2.95M	30.97	11.35 MB	18.28	336.74
SAM-Med2D 256	271.24M	92.02	2443.16 MB	48.75	1263.77

TABLE 9. Ablation results (Dice) on BTXRD at 512 × 512.

PSG	Decoder attention	Prompt	Covering	Off-center
No	None	Gaussian	0.8596	0.8230
No	None	Binary	0.8597	0.8253
No	Vanilla gate	Gaussian	0.8671	0.8408
No	CAD	Gaussian	0.8685	0.8358
Yes	None	Gaussian	0.8567	0.8297
Yes	Vanilla gate	Gaussian	0.8611	0.8314
Yes	CAD	Binary	0.8611	0.8317
Yes	CAD	Gaussian (PGA-UNet)	**0.8788**	**0.8496**

parameter count and an 11.9-fold reduction in FLOPs relative to SAM-Med2D, together with 6.0× faster GPU inference and 11.6× faster CPU inference.

I. ABLATION STUDY

The ablation study directly tests which proposed components are responsible for the observed gains. In particular, it examines whether performance comes from the prompt representation itself, from encoder-side prompt gating (PSG), from decoder-side prompt-conditioned attention (CAD), or from their combination. We isolate three factors: whether PSG is present in the encoder, what form of decoder skip-attention is used (none, the original unconditioned Attention U-Net gate, or CAD), and whether the prompt is represented as a binary box or a Gaussian-smoothed plateau. Combining these factors gives eight trained configurations per dataset, all at 512×512: the seven partial variants that isolate individual factors and their pairwise interactions, plus the full proposed PGA-UNet, which uses PSG, CAD, and the Gaussian prompt together. Tables 9 and 10 report Dice under covering and off-center prompts for BTXRD and FracAtlas, respectively.

The full PGA-UNet configuration (PSG, CAD, and the Gaussian prompt combined) is the single best row in both tables, under both covering and off-center prompts, on both datasets. This is the clearest outcome of the ablation study: none of the seven partial configurations matches the full combination in any of the four covering/off-center by dataset settings.

TABLE 10. Ablation results (Dice) on FracAtlas at 512 × 512.

PSG	Decoder attention	Prompt	Covering	Off-center
No	None	Gaussian	0.7727	0.7356
No	None	Binary	0.7670	0.7199
No	Vanilla gate	Gaussian	0.7560	0.7311
No	CAD	Gaussian	0.8158	0.7777
Yes	None	Gaussian	0.7115	0.6990
Yes	Vanilla gate	Gaussian	0.7372	0.7170
Yes	CAD	Binary	0.8132	0.7896
Yes	CAD	Gaussian (PGA-UNet)	**0.8286**	**0.7925**

The component-wise comparisons reveal a consistent pattern of synergy rather than three independent additive contributions. First, PSG in isolation, that is, without CAD in the decoder, does not reliably help and can hurt: on BTXRD it leaves Dice essentially unchanged relative to the no-PSG, no-attention baseline (0.8567 versus 0.8596 covering), while on FracAtlas it reduces Dice by 0.0612 covering and 0.0366 off-center relative to the same baseline. Encoder-side prompt amplification without a matching decoder-side mechanism to exploit it can therefore be actively counterproductive. Second, CAD in isolation, that is, without PSG, does help on both datasets (covering Dice rises by 0.0089 on BTXRD and 0.0431 on FracAtlas relative to the no-PSG, no-attention baseline), and it is more reliable than the original unconditioned attention gate: replacing CAD with the vanilla gate in the same no-PSG setting costs 0.0167 covering Dice on FracAtlas, where the unconditioned gate falls even below the no-attention baseline. Third, once PSG and CAD are combined, the pairing outperforms either component alone by a wide margin, most visibly on FracAtlas, where the full model exceeds the CAD-only variant by 0.0128 covering and 0.0148 off-center Dice, and exceeds the PSG-only variant by 0.1171 covering and 0.0935 off-center Dice. This indicates that PSG's benefit is conditional on CAD being present to make productive use of the amplified encoder features, rather than PSG being independently useful.

The prompt-representation comparison shows a related pattern. Replacing the Gaussian prompt with a binary prompt inside the full PSG+CAD architecture costs 0.0177 covering and 0.0179 off-center Dice on BTXRD, and 0.0154 covering and 0.0029 off-center Dice on FracAtlas. At the no-PSG, no-attention baseline, however, the same substitution has almost no effect (within 0.006 Dice on BTXRD, and a smaller 0.0057–0.0157 Dice gap on FracAtlas that favors Gaussian but is far below the gap observed in the full architecture). The Gaussian plateau representation therefore becomes clearly beneficial only once the network is designed to reuse prompt information throughout the encoder and decoder, again pointing to synergy between the three design choices rather than to any one of them being independently sufficient.

J. FAILURE MODES

The method still has identifiable failure cases. In the qualitative review collected during the thesis experiments, lower Dice scores were more likely for lesions with highly elongated or branching shapes, lesions located in anatomically cluttered regions such as the shoulder or clavicle, and lesions with weak contrast relative to surrounding structures. These cases are relevant because they show that prompt guidance does not eliminate ambiguity once the local image evidence itself is poor.

VI. DISCUSSION

The results support a fairly coherent interpretation of the task. First, PGA-UNet performs well on both datasets, and Monte Carlo cross-validation shows that this performance is stable

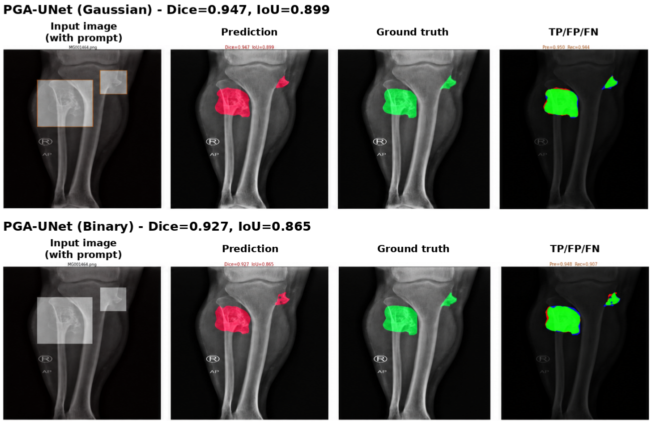

FIGURE 5. Qualitative comparison between the Gaussian prompt representation and the binary prompt representation. The Gaussian version produces cleaner boundaries and better overlap on this representative BTXRD example.

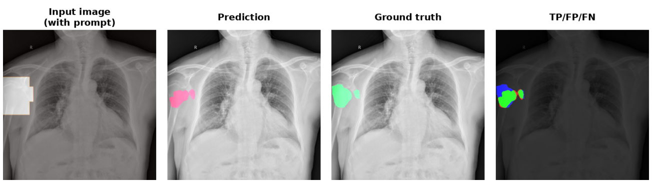

FIGURE 6. Representative failure case in an anatomically overlapping region. Even with a covering prompt, the lesion boundary remains difficult because the local X-ray structures are visually confounded.

across repeated random image-level splits rather than tied to one favorable partition. Second, Attention U-Net receives no clinician-provided box, so its result on both datasets is not a competing prompt-guided method; it instead shows how much of the difficulty in this task comes from localization alone, and the gap to PGA-UNet is largest precisely on the images where this automatic baseline is weakest. In that sense, the Attention U-Net comparison should be read as evidence for the importance of externally supplied spatial guidance, not simply as a ranking of network architectures. Third, the prompt-matched comparison that actually tests the proposed design is against fine-tuned SAM-Med2D at the matched 256×256 resolution, where PGA-UNet performs better on both datasets, under both covering and off-center prompts, with both models trained on the same 50/50 mixture of covering and off-center prompts. Fourth, this advantage becomes even clearer on the small-lesion subset, where fine-tuned SAM-Med2D nearly fails to produce a usable mask, indicating that a box prompt alone is not sufficient once the target shrinks far enough. Fifth, the ablation results show that PSG and CAD are most effective jointly rather than independently, with the full combination outperforming every partial configuration under both covering and off-center prompts on both datasets. Finally, the efficiency analysis shows that the model remains substantially lighter and faster than SAM-Med2D.

The scope of the analysis was deliberately narrowed. The auxiliary screening pipeline is excluded because it addresses a different question from prompt-guided segmentation itself. Likewise, the retained subgroup analyses are objective constructions tied either to lesion size or to a baseline-defined ranking.

The current study still has further limitations. The model depends on resizing inputs to a fixed resolution, which can dilute fine lesion detail in very small targets. Prompts are also simulated from existing annotations rather than collected from radiologists, so they do not yet capture how a clinician would actually draw a box in practice; a box drawn too tightly around the lesion is convenient experimentally but may not be realistic clinically. This matters because the central argument of the article is interactive and clinician-facing, whereas the present experiments still use a controlled prompt protocol rather than truly natural user prompts. The current formulation does not yet exploit additional information sources beyond the image and box prompt, and the training setup still requires dense mask supervision. The evaluation also remains limited to the current public datasets, with BTXRD and FracAtlas trained separately rather than jointly or evaluated through external-domain transfer. The Monte Carlo cross-validation experiments cover the main resolution and prompt-condition trend at 512×512, but the ablation study itself is still reported from a single deterministic split rather than from repeated multi-seed training.

Another limitation concerns the optimization and evaluation components. The current loss does not yet emphasize the small-lesion or accurate-center-localization properties that this task actually needs, even though CBL is already used at evaluation time. The present article therefore demonstrates the system-level effectiveness of PGA-UNet under the current loss configuration, but has not yet explored the further improvement that additional loss terms addressing small regions, lesion centers, or prediction confidence could offer.

An additional limitation is interpretive rather than purely empirical. Several observed gains, especially under prompt perturbation, are consistent with the intended roles of PSG, CAD, and the Gaussian plateau prompt, but this article does not include feature-level probing that would establish a unique internal mechanism for those gains. The ablation study therefore supports the usefulness of the design choices at the system level, not a definitive causal account of representation dynamics inside the network.

VII. CONCLUSION

This article presented PGA-UNet, a lightweight prompt-guided CNN for interactive bone X-ray lesion segmentation. The central idea is not merely to segment inside a prompted box, but to treat the box as a dense spatial prior that both tightens the search space and helps preserve weak small-lesion evidence throughout the network. By combining a plateau-style Gaussian prompt map with encoder-side Prompt Spatial Gates and decoder-side Conditional Attention Decoders, the model amplifies and preserves weak small-lesion signal rather than merely restricting where a mask should be produced. Attention U-Net, which receives no clinician-provided box, is used only to show how difficult localization is for these very small lesions; the prompt-matched comparison that actually tests the design is against fine-tuned SAM-Med2D, where PGA-UNet performs better on both BTXRD and FracAtlas while remaining substantially smaller and faster. The most consistent evidence for the design lies in the ablation study, where the full PSG+CAD+Gaussian combination outperforms every partial configuration under both covering and off-center prompts on both datasets; in the small-lesion comparison, where fine-tuned SAM-Med2D nearly fails while PGA-UNet remains usable and remains centered on the lesion according to CBL; and in greater stability on the subset of images where Attention U-Net fails, which indicates that the gain concentrates exactly where localization is hardest. Future work should evaluate the method with radiologist-provided prompts, looser and more realistic box protocols, external cohorts, higher-resolution processing without aggressive resizing, and broader clinical interaction settings.

DATA AVAILABILITY

BTXRD and FracAtlas are public datasets cited in this article. The manuscript reports experiments on image-level train/validation/test splits derived from those datasets. Code and split details should be released with the final reproducibility package.

AUTHOR CONTRIBUTIONS

Thong Luc: conceptualization, methodology, software, investigation, formal analysis, visualization, writing of the original draft. Nguyen Huu Binh: software support, data preparation, experiment verification, review and editing. Ly Quoc Ngoc: supervision, validation, review and editing.

ETHICS STATEMENT

This study used public de-identified radiograph datasets and did not involve prospective data collection, patient contact, or intervention.

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OpenAI ChatGPT was used only for vocabulary and grammar refinement in selected non-technical manuscript text. All technical content, experimental design, mathematical formulation, figures, tables, and conclusions were prepared and verified by the authors, who take full responsibility for the final manuscript.

REFERENCES

[1] A. Kirillov *et al.*, "Segment Anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 4015–4026.
[2] J. Cheng *et al.*, "SAM-Med2D," *arXiv preprint arXiv:2308.16184*, 2023.
[3] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in *MICCAI*, 2015, pp. 234–241.
[4] O. Oktay *et al.*, "Attention U-Net: Learning Where to Look for the Pancreas," in *Medical Imaging with Deep Learning*, 2018.
[5] N. Xu, B. Price, S. Cohen, J. Yang, and T. S. Huang, "Deep Interactive Object Selection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 373–381.
[6] G. Wang *et al.*, "DeepIGeoS: A Deep Interactive Geodesic Framework for Medical Image Segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 7, pp. 1559–1572, 2019.
[7] S. Yao *et al.*, "A Radiograph Dataset for the Classification, Localization, and Segmentation of Primary Bone Tumors," *Scientific Data*, vol. 12, no. 1, p. 88, 2025.
[8] I. Abedeen *et al.*, "FracAtlas: A Dataset for Fracture Classification, Localization and Segmentation of Musculoskeletal Radiographs," *Scientific Data*, vol. 10, no. 1, p. 521, 2023.
[9] J. Ma *et al.*, "Segment Anything in Medical Images," *Nat. Commun.*, vol. 15, no. 1, p. 654, 2024.

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